HEART ATTACK PREDICTION

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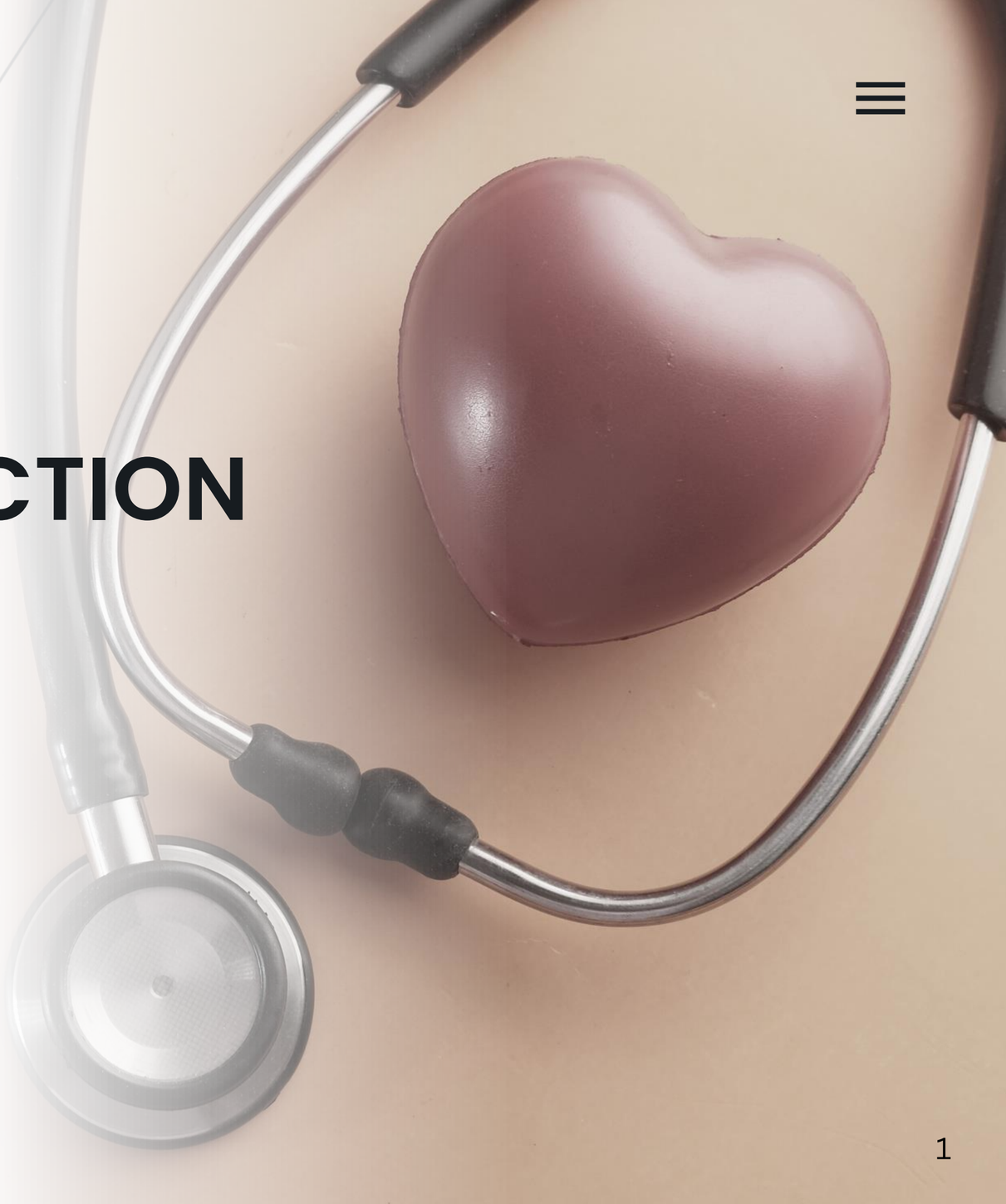

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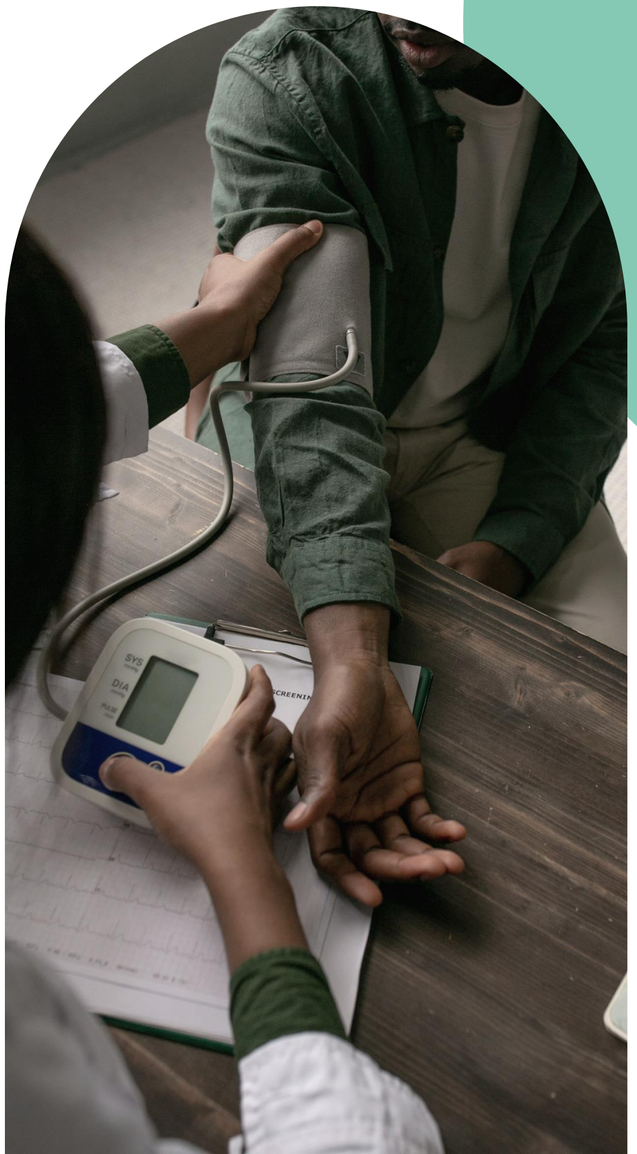
introduction

- Cardiovascular diseases are the leading cause of death globally.
- Heart attacks often occur suddenly without warning signs.
- Early prediction can save lives and reduce healthcare costs.
- Our project aims to detect heart attack risk early using big data and machine learning.



Problem Statement

- Heart attack symptoms often appear late.
- Traditional diagnosis is reactive, not predictive.
- Motivation: Reduce heart attacks using real-time, data-driven systems and Big Data technologies.
- Inspired by wearable technologies and AI advancement.

Literature Review & Research Gap

- Most existing studies rely on small, static datasets.
- There is limited use of big data frameworks in heart attack prediction.
- Real-time data streaming and analysis are rarely implemented.
- Existing solutions lack integration of wearable devices with big data pipelines.
- Our project addresses this gap by using real-time streaming tools and big data technologies for live heart attack prediction.

Prediction of Heart Disease Using Machine Learning: A Systematic Literature Review

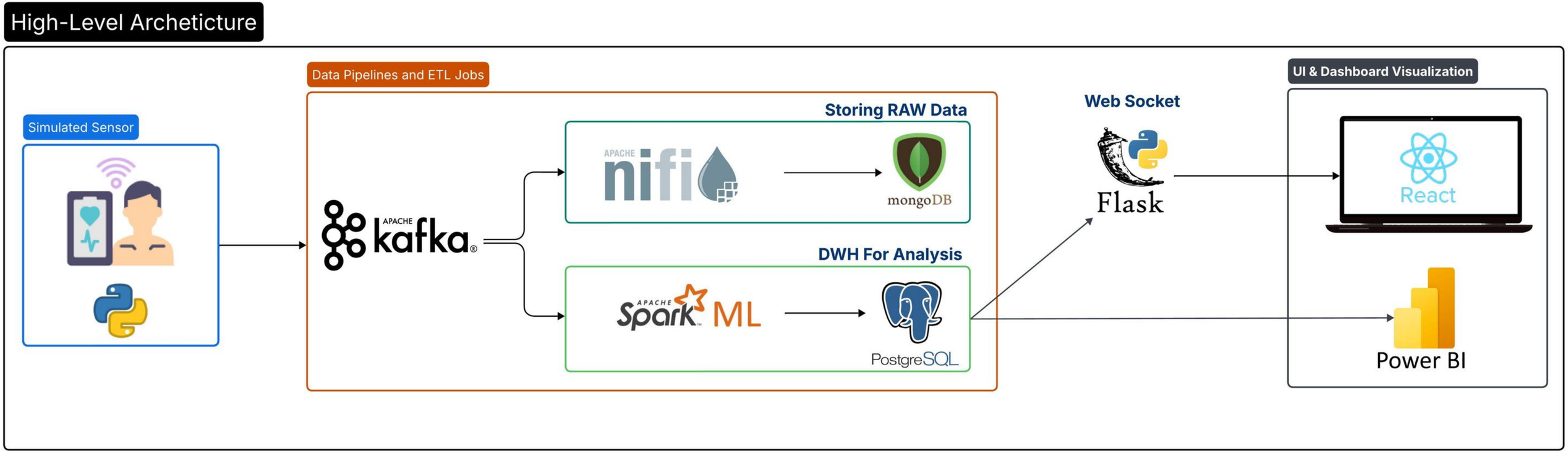
5.2. Recommendations :

Based on the large amount of information related to heart disease worldwide, it is recommended to integrate all data and apply machine learning techniques with big data. Likewise, the construction of graphical interfaces with machine learning that allow medical personnel to help better decision making is recommended.(Vergaray et al., 2023)

System Architecture & Work Flow

- Sensors→ Kafka
- Kafka → NiFi → MongoDB
- Kafka → Spark ML → PostgreSQL(DWH)
- DWH → Power Bi
- DWH → Flask (web socket)
- Flask (web socket) → React (UI & Dashboard)

System Architecture & Work Flow



System Architecture Explanation

- **Sensors → Kafka:** Collect and stream real-time health data using Apache Kafka.
- **Kafka → NiFi → MongoDB:** Use Apache NiFi to extract data from Kafka and store raw data in MongoDB for backup .
- **Kafka → Spark ML → PostgreSQL (DWH):** Spark ML processes data from Kafka, makes predictions, and stores results in a PostgreSQL data warehouse.
- **DWH → Power BI:** Power BI connects to the data warehouse to create interactive visual reports.
- **DWH → Flask (WebSocket):** Flask with WebSocket sends real-time prediction updates from the DWH.
- **Flask → React:** React frontend displays a live dashboard for users to monitor predictions and insights.

Dataset & Data Collection

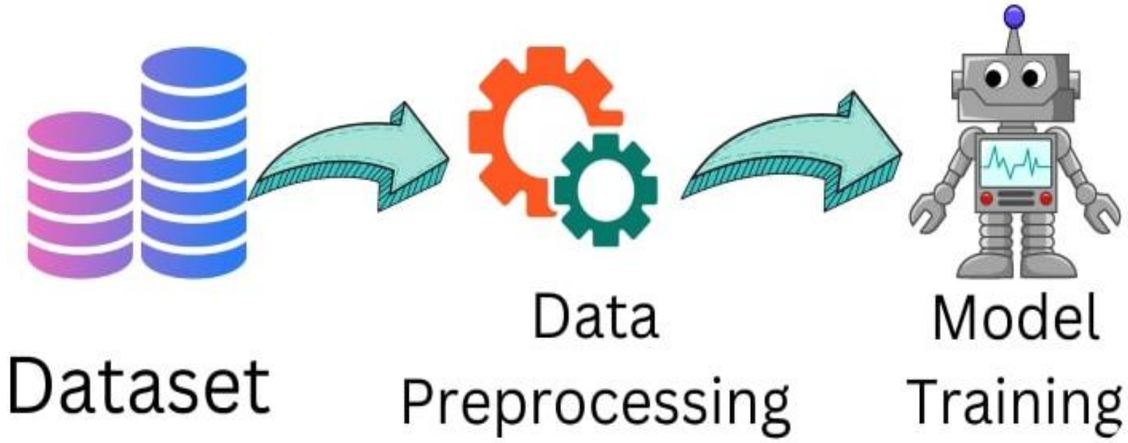
- Multiple public datasets were used to train the machine learning model (e.g., IEEE dataset, Kaggle datasets).
- These datasets include features like age, blood pressure, cholesterol, heart rate, and other clinical measurements.

GAN for Data Augmentation

- To increase our dataset size and improve model learning, we used a deep learning technique called a GAN — a Generative Adversarial Network. We trained it on our real patient data (like age, blood pressure, etc.) to generate new, similar data points. The generator creates new records, and the discriminator learns to distinguish fake from real. Over time, the generator improves and produces more realistic data. We used this method because our dataset was not big enough , and we needed more data for better training.

A Review of Machine Learning's Role in Cardiovascular Disease Prediction: Recent Advances and Future Challenges

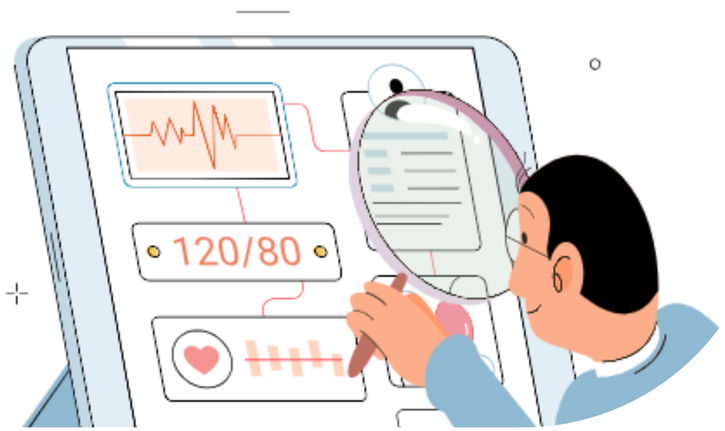
The Generative Adversarial Network (GAN) stands out as a widely embraced technique in the machine learning domain [73,213]. Leveraging this method enables the creation of synthetic data closely resembling real data, making GAN a promising solution for addressing challenges associated with data scarcity in cardiovascular discussions. (2022, 8, 239–243. [CrossRef])



Data Preprocessing

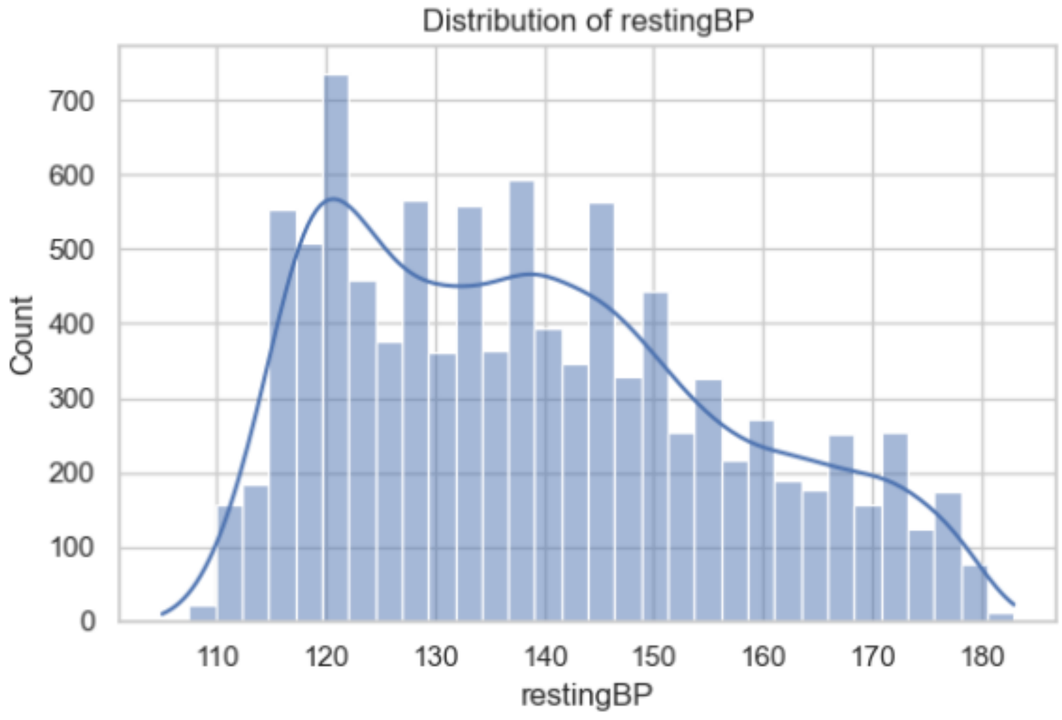
- **Check for Missing and null Values**
- **Drop Duplicates**
- **Removed outliers**

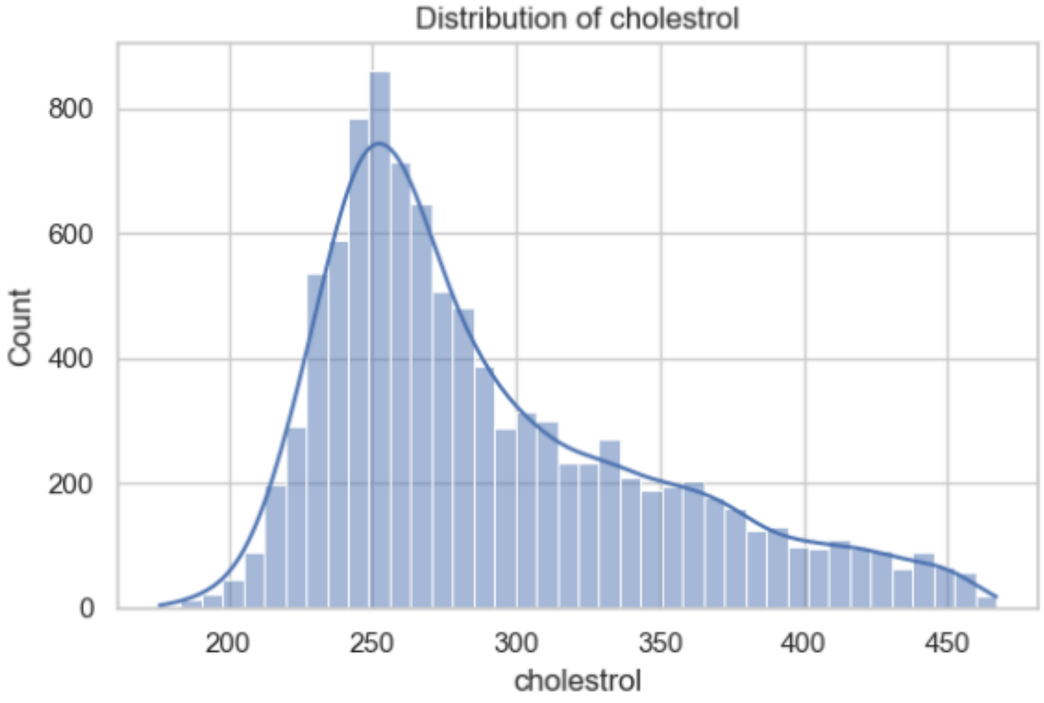
Maintaining
Patient Records

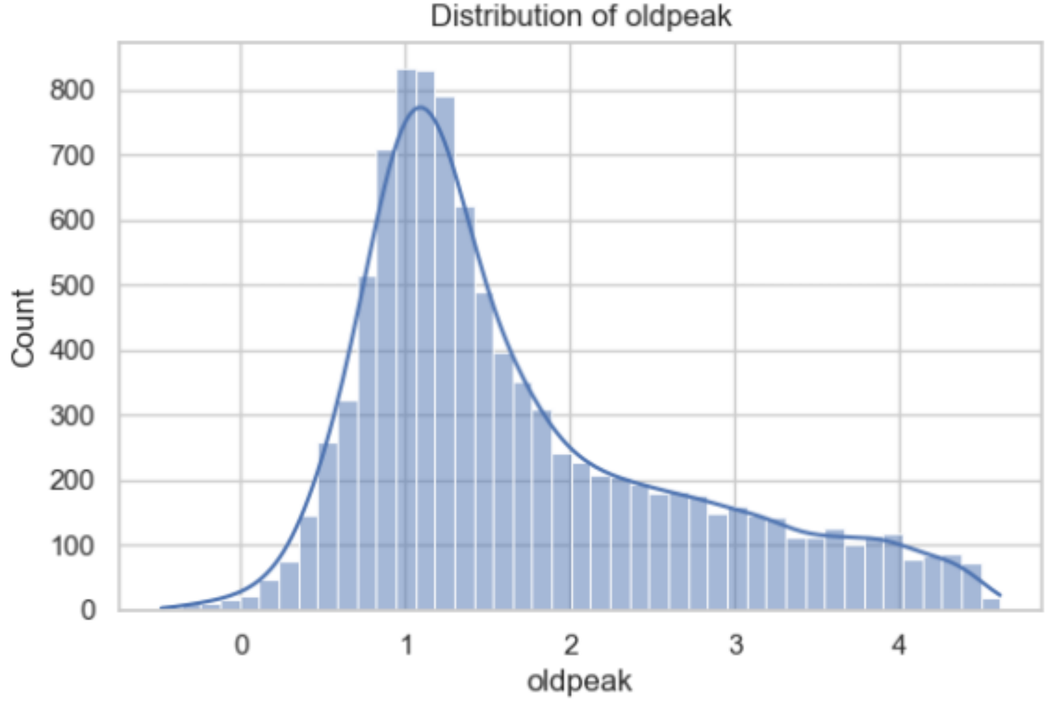


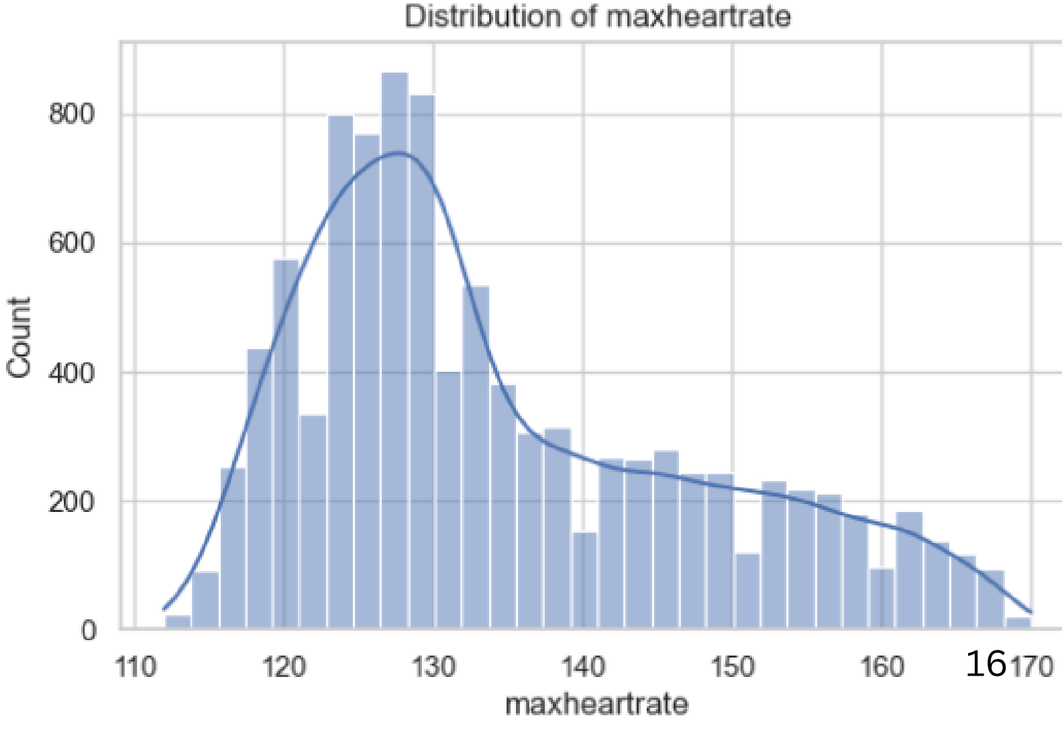
Exploratory Data Analysis (EDA)

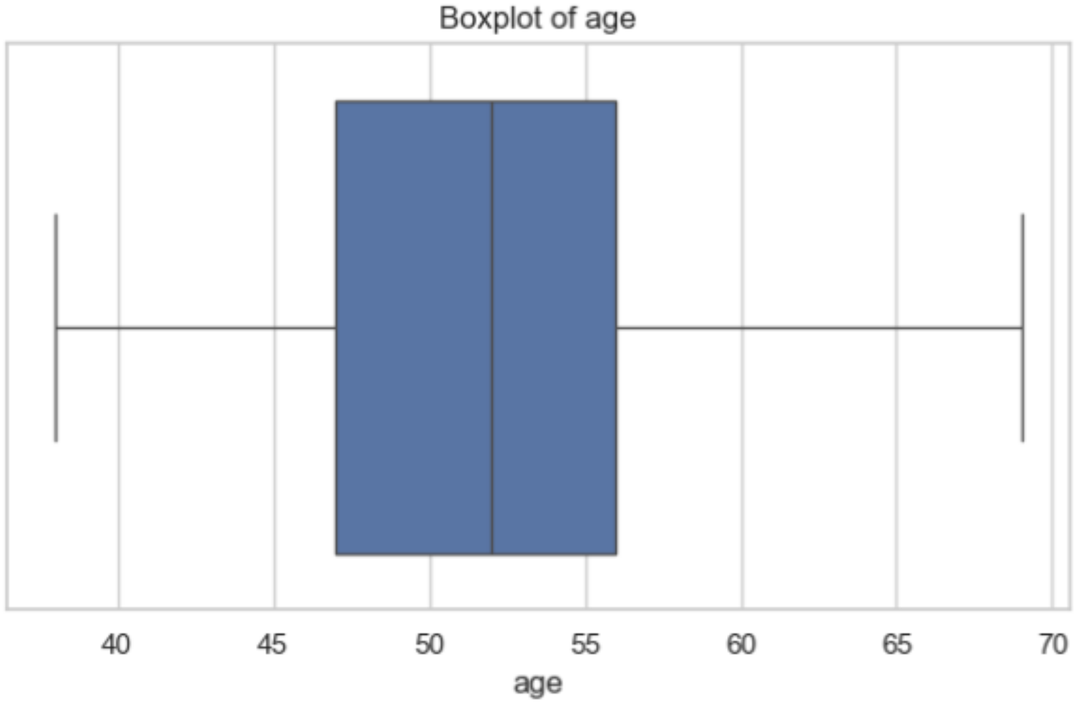
- Visualized feature distributions using histograms and box plots to detect skewness and outliers.
- Performed initial analysis on the combined training datasets to understand data structure and quality.
- Checked for missing values, duplicate records, and data consistency issues.
- Used descriptive statistics to examine distributions of key features like age, cholesterol, and heart rate.

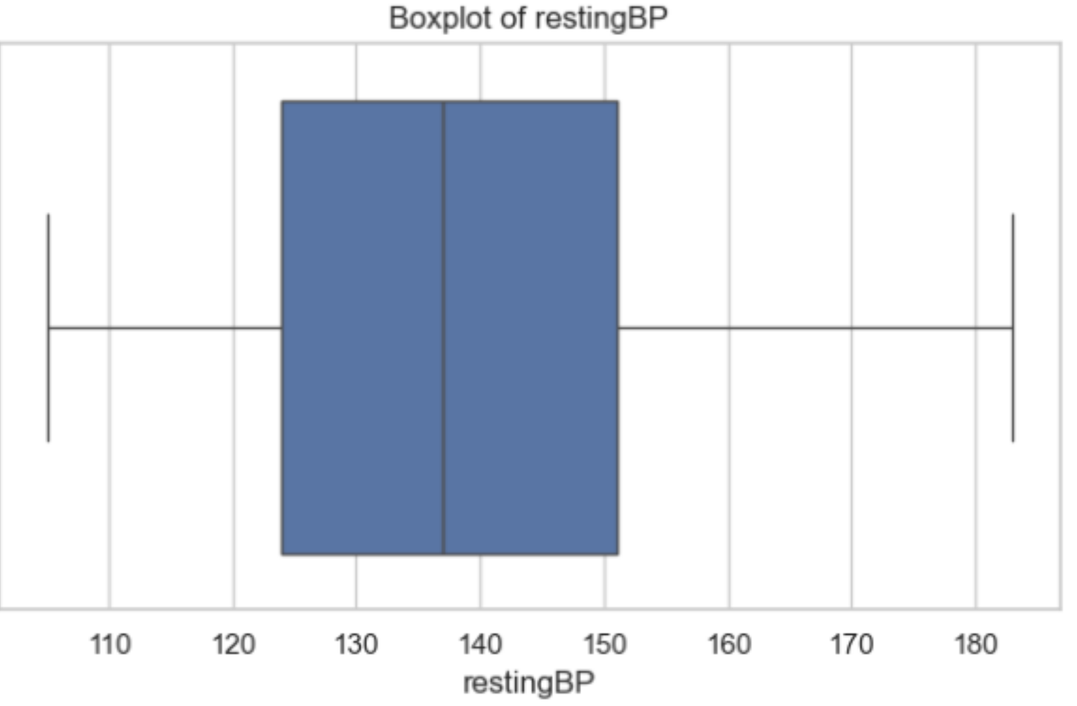

Distribution of restingBP
Count
restingBP
0
100
200
300
400
500
600
700
110
120
130
140
150
160
170
180

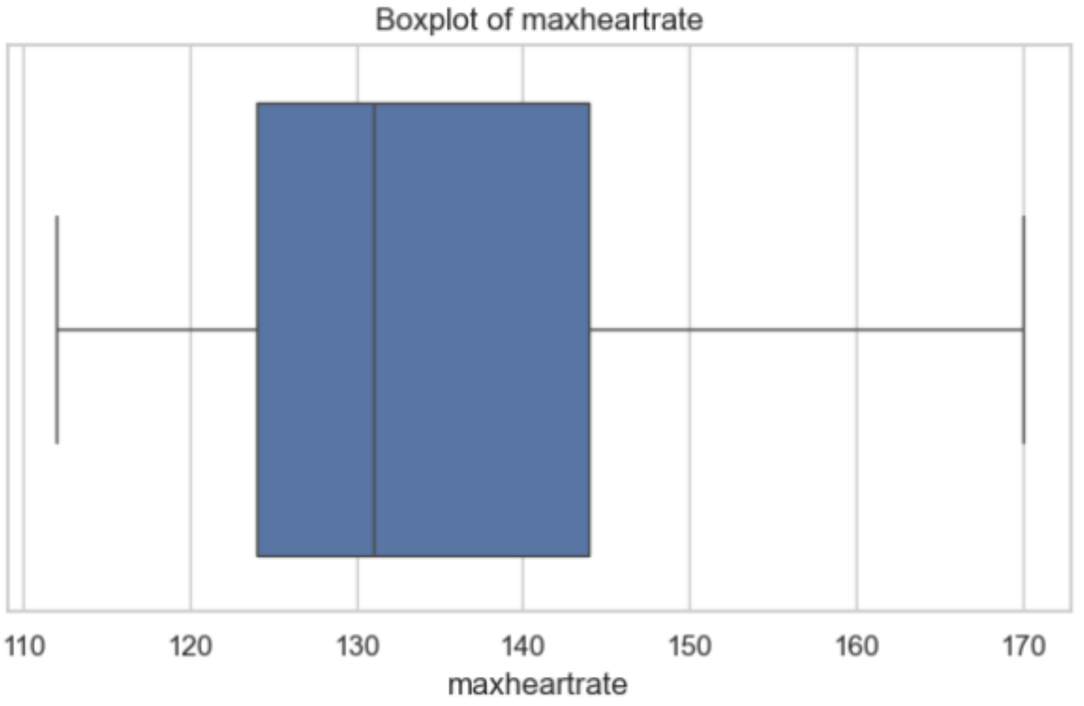

Distribution of cholestrol
Count
cholestrol
0
200
400
600
800
200
250
300
350
400
450

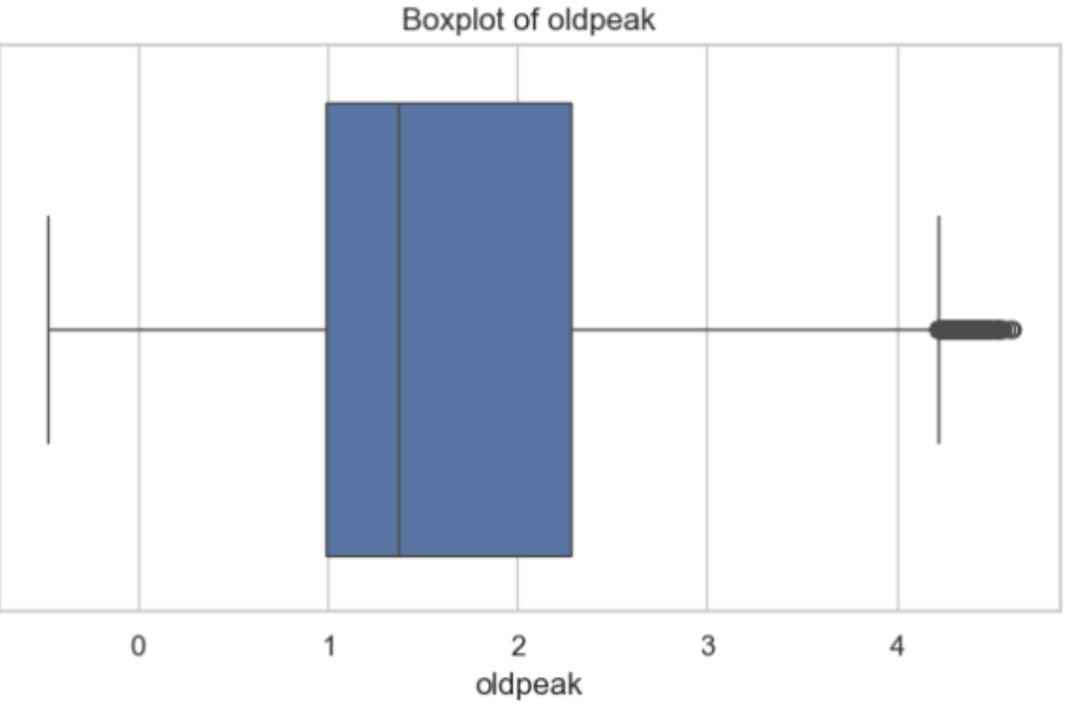

Distribution of oldpeak
Count
oldpeak
0
100
200
300
400
500
600
700
800
0
1
2
3
4

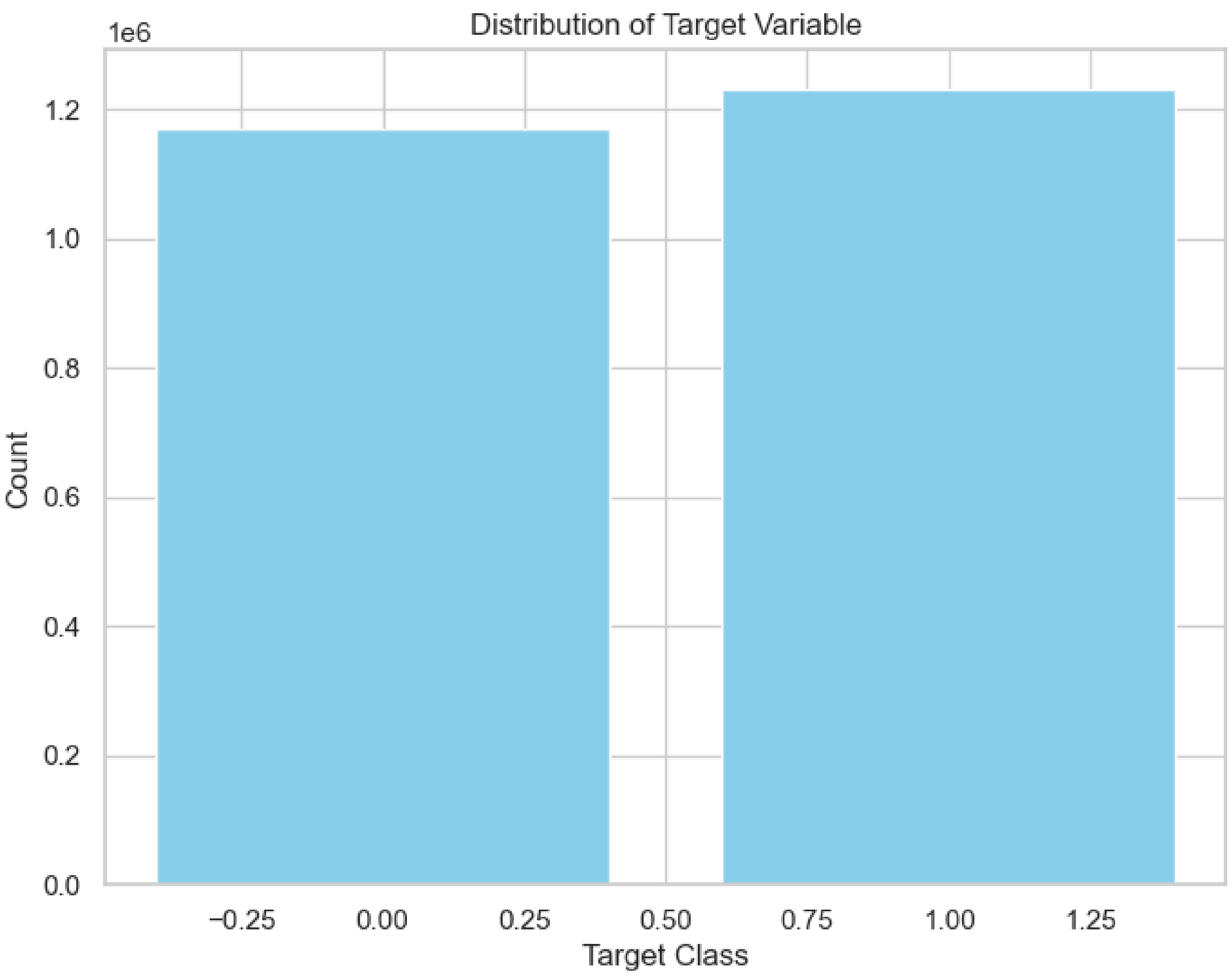

Distribution of maxheartrate
Count
maxheartrate
0
200
400
600
800
110
120
130
140
150
160
170

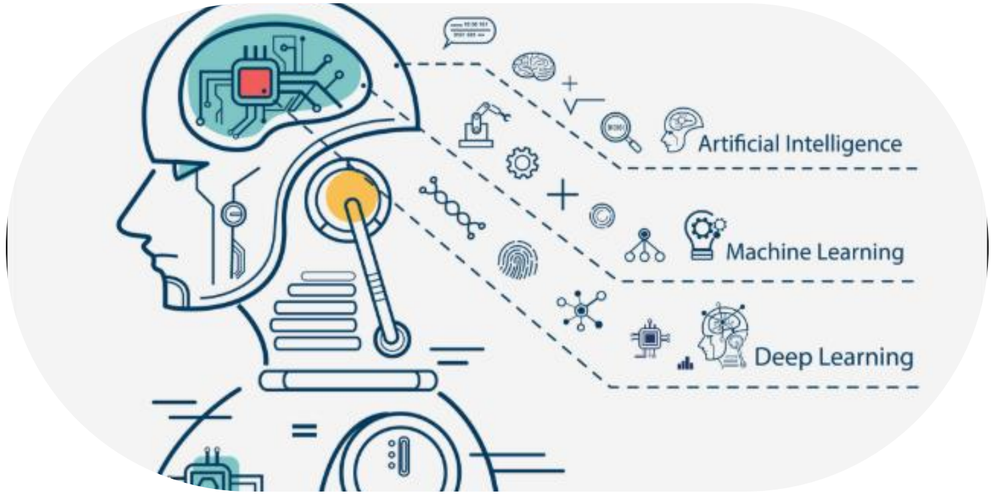

Boxplot of age
40
45
50
55
60
65
70
age


Boxplot of restingBP
110
120
130
140
150
160
170
180
restingBP


Boxplot of maxheartrate
110
120
130
140
150
160
170
maxheartrate


Boxplot of oldpeak
0
1
2
3
4
oldpeak


Distribution of Target Variable
1e6
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Count
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
Target Class



Machine Learning Model

- **Model used: Random Forest Classifier**
- **Cross-Validation**
- **Model Evaluation**
- **Deployment**

Train model

```
cvModel = crossval.fit(train)
```

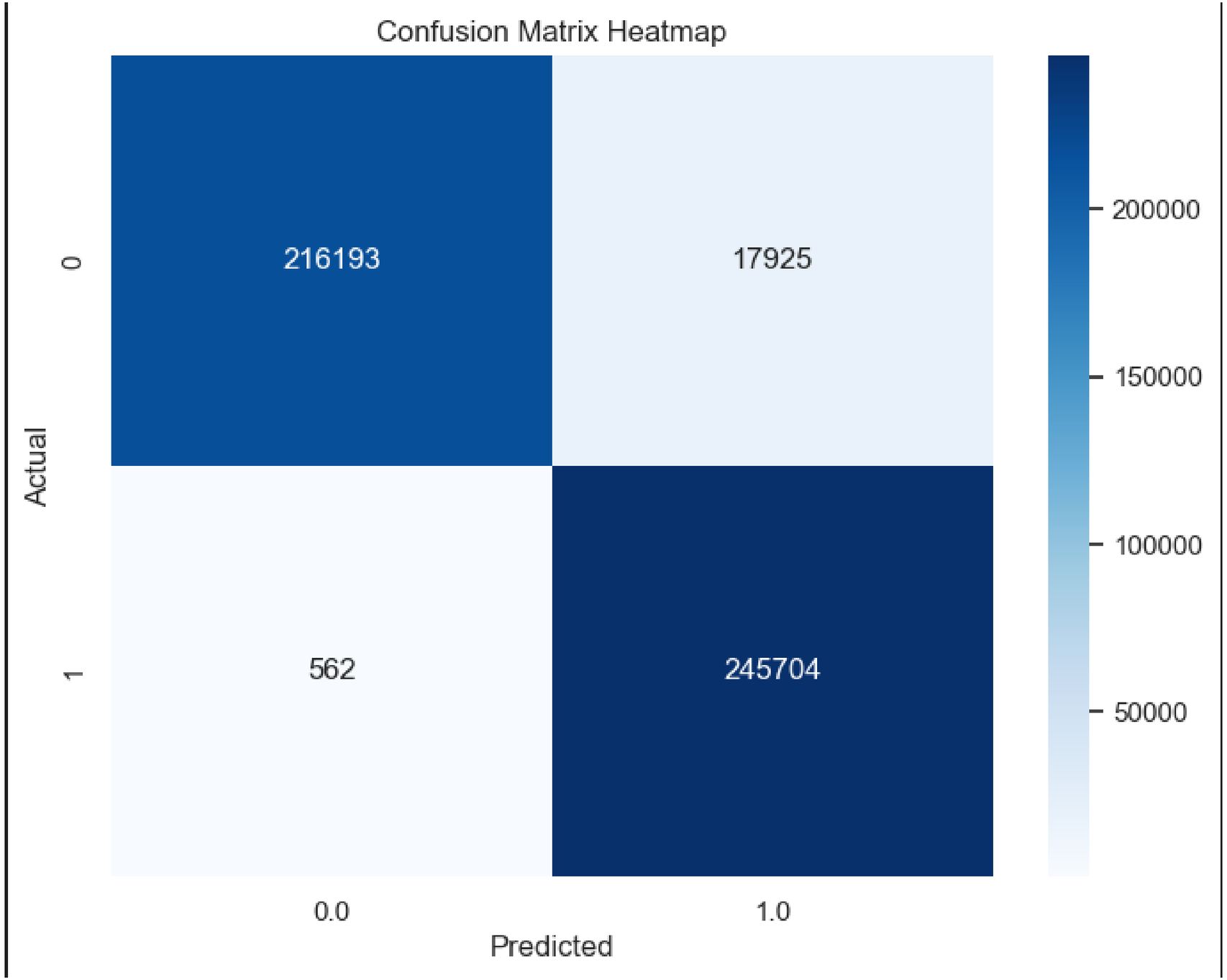
Evaluate model

```
predictions = cvModel.transform(test)
evaluator = MulticlassClassificationEvaluator(labelCol="target", predictionCol="prediction", metricName="accuracy")
accuracy = evaluator.evaluate(predictions)
print(f"Test Accuracy: {accuracy}")
```

```
Test Accuracy: 0.9615162037037037
```

The results showed that the best accuracy for cardiovascular disease classification has been achieved using a random forest algorithm with a rate of 95.4%. (33–45. CrossRef)

ACHEAVED ACCURACY : 96.2%


Confusion Matrix Heatmap
216193
17925
562
245704
0
1
Actual
0.0
1.0
Predicted
200000
150000
100000
50000

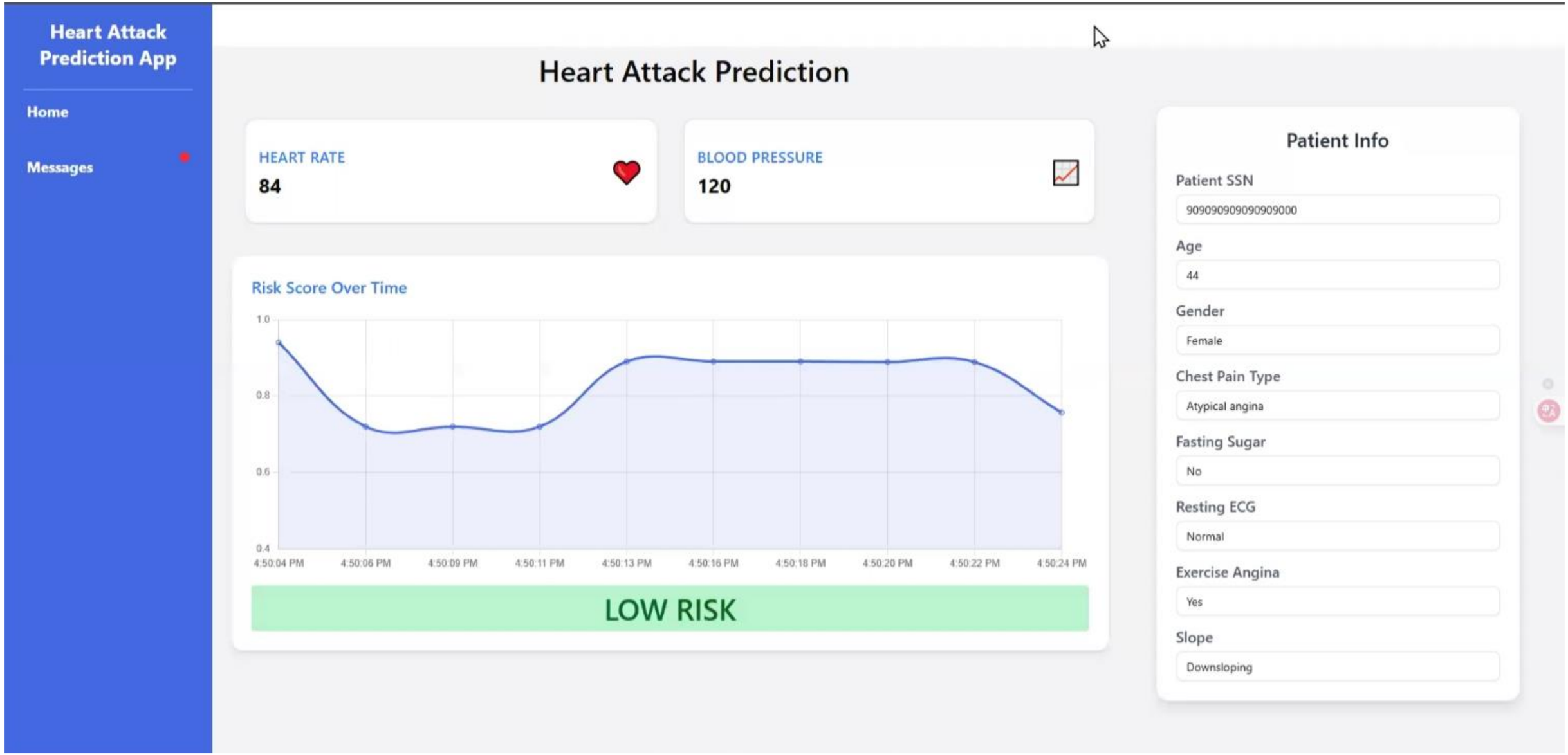
Classification Report

Class	Precision	Recall	F1-Score	Support
0 (No Risk)	0.9974	0.9234	0.9590	234,118
1 (At Risk)	0.9320	0.9977	0.9637	246,266
Accuracy			**0.9615**	480,384
Macro Avg	0.9647	0.9606	0.9614	480,384
Weighted Avg	0.9639	0.9615	0.9614	480,384

Save model

```
cvModel.bestModel.write().overwrite().save("random_forest_model")
```

UI-Visualization & Dashboard



Project Results Summary

Our system achieved 96.15% accuracy in predicting heart attack risk using real-time data streaming and machine learning. The pipeline processed simulated live data with Kafka and Spark, delivering fast and reliable predictions stored in PostgreSQL. The results confirm the effectiveness of big data tools for real-time health monitoring.

Fauget.

Conclusion & Future Work

- Successfully built an end-to-end heart attack prediction system.
- Integrated big data tools for real-time processing and ML-based prediction.
- Achieved high accuracy (96.15%) using reliable public datasets.
- Proved the potential of real-time analytics in healthcare applications.

- Integrate real sensor data from wearable devices for real-time input.
- Experiment with deep learning models to improve prediction accuracy.
- Develop a user-facing mobile or web application for monitoring.
- Collaborate with healthcare providers for clinical validation and deployment.

Thank you